MATERIAL COST

1. Meaning of Material

- The general meaning of material is all commodities/physical objects supplied to an organization to be used in producing or manufacturing of finished or intermediate goods.
- It may be classified as direct material or indirect material.

2. Control of Material

- It starts from 3Es i.e. Economy, Efficiency and Effectiveness
- In other words, it is economy in procurement, efficiency in handling and processing the material and effectiveness in producing desired output as per the standard.

3. Objective of Material Control

- **Minimized Cost** - Material should be purchased only when it is needed and in most economic quantities.
- **Availability -** Material of desired quality should be available when needed to minimize interruption in production process.
- **Lowest Purchase Price -** Purchasing of material will be made at the most favorable prices under the best possible terms.
- **Minimum Investment -** Investment in material is maintained at minimum level consistent with the operating requirement.
- **Material Storage -** Materials are, at all the time, charged as the responsibility of some individual.
- **Reduction in Wastage -** Wastage and losses while the materials are in store should be avoided as far as possible.

4. Components of Material Cost

(A) Purchase Cost = No. of units purchased × Cost per unit

(B) Ordering Cost = No. of orders × Cost per order

$$\text{No. of orders} = \frac{\text{Annual requirement}}{\text{Order Size}}$$ [Always round off to the next digit]

$$\text{Frequency of order} = \frac{365/52/12}{\text{No. of orders}}$$

(C) Carrying cost = Average quantity of goods × Carrying cost per unit per annum

$$\text{Average quantity} = \frac{\text{Order size}}{2}$$

$$\text{Average quantity with safety stock} = \text{Safety stock} + \frac{\text{Order size}}{2}$$

5. Determination of Order Size

It should be at the level where material cost is minimum.

Determination of Order Size

- Using Formula → Assume no discount exist → Objective: Minimize - OC + CC
- Using Tabular Format → When Discount exist → Objective: Minimize – PC + OC + CC

6. Economic Order Quantity (EOQ) → $\boxed{\text{At EOQ} \Rightarrow OC = CC}$

It is that order size at which sum total of ordering cost and carrying cost is minimum.

$$\boxed{EOQ = \sqrt{\frac{2 \times A \times O}{C}}}$$

Where, A = Annual requirement of raw material

O = Cost per order

C = Carrying cost per unit per annum

7. Important points for Calculation

(A) If carrying cost is given in % than such % is applied on purchase price per unit of material.

(B) If number of order is in decimal than take to the next round off number

e.g. 3.4 to 4, 3.1 to 4, 3.7 to 4 etc.

(C) A = Raw material purchased quantity (Prefer) 1 or Raw material consumed quantity 2

(I) Production units = Sale units + Closing stock FG – Opening stock FG

(II) Raw material consumption = Production units × Raw material consumption per unit

(III) Raw material purchase = RM consumption + Closing stock RM – Opening stock RM

Sale – Given
↓ Stock Adj.
Prod.
↓ Cons. P.U.
RM Cons.
↓ RM Stock Adj.
RM Purch.

PC + OC + CC = $(5000 \times Rs.20) + (25 \times Rs.16) + \left(\frac{200}{2} \times Rs.20 \times 20\%\right)$ = Rs.100800

Question – 1

(i) Compute EOQ and total variable cost for the following:

Annual Demand (A)	=	5,000 units
Unit price	=	₹20
Order cost (O)	=	₹16
Storage cost (C)	=	2% per annum
Interest rate (C)	=	12% per annum
Obsolescence rate (C)	=	6% per annum

20% p.a.

$\sqrt{\frac{2 \times 5000 \times 16}{20\% \times 20}}$ = 200 units

No. of orders = $\frac{5000}{200}$ = 25

$\sqrt{\frac{2 \times 5000 \times 16}{20\% \times 12.80}}$ = 250 units

No. of orders = $\frac{5000}{250}$ = 20

(ii) Determine the total cost that would result for the items if a new price of ₹12.80 is used.

[Answer – (i) 200 units; ₹1,00,800; (ii) ₹64,640]

↳ $(5000 \times 12.80) + (20 \times Rs.16) + \left(\frac{250}{2} \times \frac{20}{100} \times 12.8\right)$ = Rs. 64640

Question – 2

SK Ltd. has received an offer of quantity discounts on its order of materials as under:

(OC + CC + PC) TC

No. of order	OS	Tons (No.)	OC	CC	Price per tons (₹)	PC
20	200	Less than 250	120	$\frac{200}{2} \times \frac{20}{100} \times 6$	6.00	24000
16	250	250 and less than 800	96	$\frac{250}{2} \times \frac{20}{100} \times 5.9$	5.90	23600
5	800	800 and less than 2,000	30		5.80	23200
2	2000	2,000 and less than 4,000	12		5.70	
1	4000	4,000 and above	6		5.60	

Lowest

The annual requirement for the materials is 4,000 tons. The ordering cost per order is ₹6 and the carrying cost is estimated at 20% per annum. You are required to compute the most Economic Order Quantity presenting the relevant information in a tabular form.

Solution

Tons	Price	Order Size	Purchase Cost	Ordering Cost	Carrying Cost	Total Cost
Less than 250	6	200	4,000×6 = 24,000	$\frac{4000}{200} \times 6 = 120$	$\frac{200}{2} \times 20\% \times 6 = 120$	24,240
250 to 800	5.90	250	4,000×5.90 = 23,600	$\frac{4000}{250} \times 6 = 96$	$\frac{250}{2} \times 20\% \times 5.9 = 148$	23,844
800 to 2,000	5.80	800	4,000×5.80 = 23,200	$\frac{4000}{800} \times 6 = 30$	$\frac{800}{2} \times 20\% \times 5.8 = 464$	23,694
2,000 to 4,000	5.70	2,000	4,000×5.70 = 22,800	$\frac{4000}{2000} \times 6 = 12$	$\frac{2000}{2} \times 20\% \times 5.7 = 1,140$	23,952
4,000 & above	5.60	4,000	4,000×5.60 = 22,400	$\frac{4000}{4000} \times 6 = 6$	$\frac{4000}{2} \times 20\% \times 5.6 = 2,240$	24,646

Total cost is lowest at order size of 800. So, economic order quantity is 800 units.

$A = 16000 \times 4 \times 6 = 3.84L$ kg $O =$ Rs.2000 $C = \frac{15}{100} \times 40 =$ Rs.6 $EOQ = \sqrt{\frac{2 \times 3.84L \times 2000}{6}}$ $= 16000$

Question – 3	

Ani Ltd. uses 6 kg. of Material 'EXE' to produce 1 finished unit of Product 'EME'. The current demand of Product 'EME' is 16,000 units quarterly. 1 kg. of Material 'EXE' costs ₹40. The cost relating to quotations, documentation works, employee cost directly attributable to the procurement of material, every-time the order is made, is ₹2,000. The cost of fund invested in inventories, cost of storage, insurance cost, etc. is estimated to be 15% per annum of average inventory.

You are required the following:

(a) Calculate the Economic Order Quantity for Material 'EXE'.

(b) Comment, should Ani Ltd. accept an offer of 2.5% discount by the supplier of Material 'EXE', if supply of the annual requirement of the Material is made in 4 equal instalments? → $\frac{3.84L}{4} = 96000$

Solution

	OS = 16000	OS = 96000
No. of order	3.84L ÷ 16000 = 24	4
PC	15360000	14976000
OC	48000	8000
CC	$\frac{16000}{2} \times \frac{15}{100} \times 40 = 48000$	$\frac{96000}{2} \times \frac{15}{100} \times 39 = 280800$
TC		

(i) EQO = $\sqrt{\frac{2 \times A \times O}{C}}$

A = 16,000 × 4 × 6 = 3,84,000 kg

O = ₹2,000

C = ₹40 per unit × 15% = ₹6 per kg

EOQ = $\sqrt{\frac{2 \times 3,84,000 \times 2,000}{6}}$ = 16,000 kg

(ii) Statement showing evaluation of proposal

Proposed order size = 3,84,000 ÷ 4 = 96,000 kg

Particulars	Order 16,000 kg	Order 96,000 kg
Number of orders	$\frac{3,84,000}{16,000} = 24$	$\frac{3,84,000}{96,000} = 4$
Annual purchase cost (₹40/39 p.u)	1,53,60,000	1,49,76,000
Annual Ordering cost (₹2,000 per order)	48,000	8,000
Annual carrying cost (₹6/5.85 per unit)	48,000	2,80,800
Total Cost	₹1,54,56,000	₹1,52,64,800

Since the total cost is lower by ₹1,91,200 in case when the company gets the discount offer of 2.5%, thus, it is recommended to accept the discount offer with order size of 96,000 kg.

Question – 4	

Catalist Ltd. is a distributor of industrial chemicals, providing the chemical in drum packaging.

Each drum of the chemical costs ₹200 from a supplier and the company sells it for ₹240.

Annual demand is estimated to be for 2,50,000 drums.

The cost of delivery is estimated at ₹100 per order and the annual variable holding cost per drum at ₹180 plus 10% of purchase cost.

$180 + \left(\frac{10}{100} \times 200\right) = 200$ $\sqrt{\frac{2 \times 2,50000 \times 100}{200}} = 500$

Based on above data, the managing director calculates the economic order quantity and suggests that it should serve as the foundation for purchasing decisions in upcoming periods.

10%×{2L - OC - CC}

However, the purchasing manager states that the managing director has ignored his share of bonus of 10% of the amount by which total annual inventory holding and order costs (before such remuneration) are below ₹2,00,000. He further points out that the suppliers also offer quantity discounts on purchase orders, i.e. if the order size is 1,000 drums or above, the price per drum is only ₹199.60, compared to ₹200 when an order is between 500 and 999 drums.

You are required to:

(a) Calculate the economic order quantity as calculated by the company's managing director. 500

(b) Comment whether Catalist Ltd. can look forward to the quantity discount offered for purchasing 1,000 drums, after calculating total cost considering purchasing manager's bonus along with supplier quantity discounts.

Solution

(a) EQO = $\sqrt{\frac{2 \times A \times O}{C}}$

A = 25,000 drums

O = ₹100

C = ₹180 + (10% × ₹200) = ₹200 per kg

EOQ = $\sqrt{\frac{2 \times 25{,}000 \times 100}{200}}$ = 500 kg

(b) Statement showing evaluation of proposal

Particulars	Order 500 drums	Order 1,000 drums
Number of orders	$\frac{2{,}50{,}000}{500}$ = 500	$\frac{2{,}50{,}000}{1{,}000}$ = 250
Annual purchase cost	2,50,000 × 200 = 5,00,00,000	2,50,000 × 199.6 = 4,99,00,000
Annual Ordering cost	500 × 100 = 50,000	250 × 100 = 25,000
Annual carrying cost	$\frac{500}{2}$ × [180 + (10% × 200)] = 50,000	$\frac{1000}{2}$ × [180 + (10% × 199.6)] = 99,980
Bonus	[10% × (2,00,000 – 50,000 – 50,000)] = 10,000	[10% × (2,00,000 – 25,000 – 99,980)] = 7,502
Total Cost	5,01,10,000	5,00,32,482

Since the total cost is decreasing, thus recommended to accept the proposal.

$O = 360 + 390 = 750$ $C = (0.50 \times 12) + 9 = 15$ $A = \frac{1}{2.5} \times 100000 = 40000$

Question – 5

A company manufactures a product from a raw material which is purchased at ₹60 per kg. The company incurs a handling cost of ₹360 plus freight of ₹390 per order. The incremental carrying cost of inventory of raw material is ₹0.50 per kg per month. In addition, the cost of working capital finance on the investment in inventory of raw material is ₹9 per kg per annum. The annual production of the product is 1,00,000 units and 2.5 units are obtained from one kg of raw material.

(FW)

Required:

$\sqrt{\frac{2 \times 40000 \times 750}{15}} = 2000$ No. of order $= \frac{40000}{2000} = 20$

(a) Calculate the economic order quantity of raw material

Freq. $= \frac{360}{20} = 18$ days

(b) Advise, how frequently should orders for procurement be placed. (Assuming 360days in the year)

$\frac{40000}{4} = 10000$ (OS)

(c) If the company proposes to rationalize placement of orders on quarterly basis, what percentage of discount in the price of raw materials should be negotiated?

Let New PP = y

	OS = 2000	OS = 10000
PC	2400000	40000y
OC	15000	3000
CC	$\frac{2000}{2} \times 15 = 15000$	$\frac{10000}{2} \times 15 = 75000$
TC	2430000	40000y + 78000

$40000y + 78000 = 2430000$

$y = 58.80$

Disc. $= 60 - 58.80 = 1.20$

Disc % $= \frac{1.20}{60} \times 100 = 2\%$

Solution

(a) A = 1,00,000 ÷ 2.5 = 40,000 kg

O = 360 + 390 = ₹750

C = 9 + (0.5 × 12) = ₹15

$$\text{EOQ} = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 40{,}000 \times 750}{15}} = 2{,}000 \text{ kg}$$

(b) Number of orders to be placed $= \frac{\text{Annual requriement of material}}{\text{Order size (EOQ)}} = \frac{40{,}000}{2{,}000} = 20$ orders

Frequency of order $= \frac{360}{No.\ of\ orders} = \frac{360}{20} = 18$ days

(c) Desired number of orders = 4

$\therefore$ desired order size $= \frac{40{,}000}{4} = 10{,}000$ kg

Let new price = y

Statement of Cost

Costs	Order Size = 2,000	Order Size = 10,000
Purchase Cost	40,000 × 60 = 24,00,000	40,000 × y = 40,000y
Ordering Cost	$\frac{40{,}000}{2{,}000} \times 750 = 15{,}000$	$\frac{40{,}000}{10{,}000} \times 750 = 3{,}000$
Carrying Cost	$\frac{2{,}000}{2} \times 15 = 15{,}000$	$\frac{10{,}000}{2} \times 15 = 75{,}00$
Total Cost	24,30,000	40,000y + 78,000

Now to rationalize cost of both options, total cost should be same under both options.

$\therefore$ 24,30,000 = 40,000y + 78,000

y = ₹58.80

$\therefore$ Discount per unit = ₹60 – ₹58.80 = ₹1.20

Discount % $= \frac{1.20}{60} \times 100 = 2\%$

No. of order $= \frac{4000}{300} = 13.33$

$EOQ = \sqrt{\frac{2 \times 4000 \times 135}{12}} = 300$ Rel. Cost = OC + CC = $(14 \times 135) + (\frac{300}{2} \times 12)$ = Rs. 3690

Question – 6

The annual demand for an item of raw material is 4,000 units and the purchase price is expected to be ₹90 per unit. The incremental cost of processing an order is ₹135 and the annual cost of storage is estimated to be ₹12 per unit. Compute the optimal order quantity and total relevant cost of this order quantity?

New $EOQ = \sqrt{\frac{2 \times 4000 \times 80}{12}} = 231$ No. of orders $= \frac{4000}{231} = 17.32 \approx 18$

Suppose that ₹135 as estimated to be the incremental cost of processing an order is incorrect and should have been ₹80. All other estimates are correct. Estimate the difference in cost on account of this error?

Assume at the commencement of the period that a supplier offers 4,000 units at a price of ₹86. The materials will be delivered immediately and placed in the stores. Assume that the incremental cost of placing the order is zero and original estimate of ₹135 for placing an order for the economic batch is correct. Analyze, should the order be accepted?

Solution

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 4{,}000 \times 135}{12}} = 300 \text{ units}$$

Number of orders = 4,000 ÷ 300 = 13.33 or 14 orders

Relevant cost of this order quantity:

Ordering cost [14 × 135]	1,890
Carrying cost [(300 ÷ 2) × 12]	1,800
Relevant cost	3,690

$$\text{Revised } EOQ = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 4{,}000 \times 80}{12}} = 231 \text{ units}$$

Number of orders = 4,000 ÷ 231 = 17.31 or 18 orders

Statement of Cost

Costs	Order Size = 300	Order Size = 231
Ordering Cost	14 × 80 = 1,120	18 × 80 = 1,440
Carrying Cost	$\frac{300}{2} \times 12 = 1{,}800$	$\frac{231}{2} \times 12 = 1{,}386$
Total Cost	2,920	2,826

Difference in cost on account of this error = 2,920 – 2,826 = ₹94

Statement of Evaluation of Offer

Costs	Order Size = 300	Order Size = 4,000
Purchase Cost	4,000 × 90 = 3,60,000	4,000 × 86 = 3,44,000
Ordering Cost	14 × 135 = 1,890	$\frac{4{,}000}{4{,}000} \times 0 = 0$
Carrying Cost	$\frac{300}{2} \times 12 = 1{,}800$	$\frac{4{,}000}{2} \times 12 = 24{,}000$
Total Cost	3,63,690	3,68,000

This special offer at ₹86 per unit should not be accepted as its total cost is higher as compared to original offer.

Prod. = 5l + 0 − 12000 = 4.88l → Cons. of P = 4.88l × 4 = 19.52l → Purch. = 19.52l + 0 − 0.24l = 19.28l
Cons. of M = 4.88l × 16 = 78.08l → Purch. = 78.08l + 0 − 0.52l = 77.56l

Question – 7	

SK Ltd. manufactures a product S which requires two raw materials P and M in a ratio of 1:4. The sales department has estimated a demand of 5,00,000 units for the product for the year. To produce one unit of finished product, 4 units of material P is required.

EOQ P = $\sqrt{\frac{2 \times 19.28l \times 15000}{13\% \times 150}}$ =

EOQ M = $\sqrt{\frac{2 \times 77.56l \times 15000}{13\% \times 200}}$ =

Stock position at the beginning (of) of the year is as below:

Product SK FG → 12,000 units

Material P } RM → 24,000 units

Material M → 52,000 units

To place an order the company has to spend ₹15,000. (O) The company is financing its working capital using a bank cash credit @ 13% (C) p.a. Product SK is sold at ₹1,040 per unit. Material P and M are purchased at ₹150 and ₹200 respectively.

A = 19.28l + 77.56l = 96.84l

O = Rs. 15000

C = $\frac{13}{100} \times \left(\frac{(19.28l \times 150) + (77.56l \times 200)}{96.84l}\right)$

Required: Compute economic order quantity (EOQ):

(a) If purchase order for both materials is placed separately

(b) If purchase order for both materials is not placed separately

Solution

+ Cl. St.

Annual production of Product SK = Annual demand – Opening stock = 5,00,000 – 12,000 = 4,88,000 units

Annual requirement of raw material = (Annual Production × Material per unit) – Opening stock + Cl.

Material P = (4,88,000 × 4) – 24,000 = 19,28,000 units

Material M = (4,88,000 × 16) – 52,000 = 77,56,000 units

(a) EOQ of Material P = $\sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 19{,}28{,}000 \times 15{,}000}{13\% \times 150}}$ = 54,462 units

EOQ of Material M = $\sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 77{,}56{,}000 \times 15{,}000}{13\% \times 200}}$ = 94,600 units

(b) EOQ of Material P & M Combined = $\sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times (19{,}28{,}000 + 77{,}56{,}000) \times 15{,}000}{13\% \times 190^*}}$ = 1,08,452 units

Material P quantity = $\frac{1{,}08{,}452 \times 19{,}28{,}000}{96{,}84{,}000}$ = 21,592 units

Material M quantity = $\frac{1{,}08{,}452 \times 77{,}56{,}000}{96{,}84{,}000}$ = 86,860 units

*Price = $\frac{(150 \times 19{,}28{,}000) + (200 \times 77{,}56{,}000)}{(19{,}28{,}000 + 77{,}56{,}000)}$ = ₹190

W. Avg

8. Inventory Control

- Its main objective is to maintain a trade-off between stock-out and over-stocking.
- It can be done on following basis:
  - (a) By setting quantitative levels
  - (b) On the basis of relative classification
  - (c) Using ratio analysis
  - (d) Physical control

9. Inventory Control

It is done by setting various levels of stock which are as follows:

- (i) Re-order quantity or EOQ → Qty. to be ordered at a point of time
- (ii) Re-order level
- (iii) Maximum level
- (iv) Minimum level
- (v) Average level
- (vi) Danger level
- (vii) Buffer stock

10. Levels of Inventory

(A) Re-Order Level (ROL) - It is the level at which purchase manager will issue a fresh purchase order with supplier.

Re-order level (ROL) = Maximum consumption × Maximum lead time
= Safety stock + (Average consumption × Average lead time)
= Minimum stock + (Average cons. × Average lead time)

(B) Maximum Level - It is the level beyond which goods are not allowed to exceed. It is fixed to avoid the costs of over-stocking.

Maximum level = ROL + ROQ – (Minimum cons. × Minimum lead time)

(C) Minimum level - It is the lowest quantity of a particular material which must be held in the store at all times. It is the level of goods at which the fresh goods ordered earlier should reach the factory premises.

Minimum level = ROL – (Average consumption × Average lead time)

(D) Average Level - It is the average quantity of goods held in the stores.

Average level $= \frac{Minimum\ level + Maximum\ level}{2}$

Average level = Minimum level + $\frac{Re-Order\ quantity}{2}$

(E) Danger Level - Danger level is the level at which normal issues of the raw material inventory are stopped and emergency issues are only made on special requisition approved by the competent authority.

Danger level = Average consumption × Emergency lead time
= Minimum consumption × Emergency lead time

(F) Buffer Stock – This stock is kept for contingency and to be used in case of sudden order.

Question – 8	

A company uses three raw materials A, B and C for a particular product for which the following data apply:

Raw material	Usage per unit (Kg)	Reorder Quantity (Kg)	Price per Kg (₹)	Delivery period Min	Average	Max	Reorder level (Kg)	Minimum level (Kg)
A	10	10,000	0.10	1	2	3	8,000	
B	4	5,000	0.30	3	4	5	4,750	
C	6	10,000	0.15	2	3	4		2,000

FG

Weekly production varies from 175 to 225 units, averaging 200 units of the said product. What would be the following quantities:

(a) Minimum stock of A? → ROL − (Avg. × Avg.) = 8000 − (200 × 10 × 2) = 4000

(b) Maximum stock of B? → ROL + ROQ − (Min. × Min.) = 4750 + 5000 − (175 × 4 × 3) =

(c) Re-order level C? → Max. × Max. = 225 × 6 × 4 = 5400 (or) 2000 + (200 × 6 × 3) = 5600

(d) Average stock level of A? → Min. + $\frac{ROQ}{2}$ = 4000 + $\frac{10000}{2}$ = 9000

Solution

(a) Minimum stock of A = ROL – (Average lead time × Average consumption)
= 8,000 – (2× 200 × 10) = 4,000 kg

(b) Maximum stock of B = ROL + ROQ – (Min. lead time × Min. consumption)
= 4,750 + 5,000 – (3× 175 × 4) = 7,650 kg

(c) Re-order level of C = Max. lead time × Max. consumption
= 4 × 225 × 6 = 5,400 kg

(d) Average level of A = Minimum level + $\frac{ROQ}{2}$ = 4,000 + $\frac{10,000}{2}$ = 9,000 kg

Question – 9

M/s SK Ltd. are the manufacturers of picture tubes for T.V. The following are the details of their operation during the year:

Average monthly market demand → FG	2,000 tubes
Ordering cost	₹100 per order
Inventory carrying cost	20% per annum
Cost of tubes	₹500 per tube
Normal usage (RM)	100 tubes per week ×52 = 5200
Minimum usage	50 tubes per week
Maximum usage	200 tubes per week
Lead time to supply	6-8 weeks → Avg. = 7

Compute the following from the above information:

(a) Economic order quantity. If the supplier is willing to supply quarterly 1,500 units at a discount of 5%, is it worth accepting? yes

(b) Maximum level of stock → 1600 + 102 − (6×50) =

(c) Minimum level of stock → 1600 − (100× 7) =

(d) Reorder level → Max. × Max. = 200 × 8 = 1600

Solution

(a) A = 100 × 52 = 5,200 (We have to consider only consumption and not monthly demand)

O = ₹100

C = 20% × 500 = ₹100

$$EOQ = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 5{,}200 \times 100}{100}} = 102 \text{ tubes}$$

Statement of Evaluation of Offer

Costs	Order Size = 102	Order Size = 1,500
Purchase Cost	5,200 × 500 = 26,00,000	5,200 × (500-5%) = 24,70,000
Ordering Cost	$\frac{5{,}200}{102} = 50.98$ *or* $51 \times 100 = 5{,}100$	$\frac{5{,}200}{1{,}500} = 3.47$ *or* $4 \times 100 = 400$
Carrying Cost	$\frac{102}{2} \times 20\% \times 500 = 5{,}100$	$\frac{1{,}500}{2} \times 20\% \times 475 = 71{,}250$
Total Cost	26,10,200	25,41,650

Since the total cost is lower at order size of 1,500, thus it is recommended to accept the offer.

(b) Re-order level = Max. consumption × Max. lead time

= 200 × 8 = 1,600 tubes

(c) Minimum level = ROL – (Avg. lead time × Avg. Consumption)

= 1,600 – (100 × 7) = 900 tubes

(d) Maximum level = ROL + ROQ – (Minimum consumption × Minimum lead time)

= 1,600 + 102 – (50 × 6) = 1,402 tubes

Present OS = 6250 $A = 6250 \times \frac{12}{3} = 25000$

Question – 10

A company buys in lots of 6,250 units which is a 3 month's supply. The cost per unit is ₹2.40. Each order costs ₹45 and inventory carrying cost is 15% of average inventory value.

Required:

(a) What is the total annual cost of existing inventory policy?

(b) How much money could be saved by employing the economic order quantity?

(c) If the company operates 250 days a year, the procurement time is 10 days and safety stock is 500 units. Find the reorder level, maximum level, minimum level and average inventory level.

$500 + \left(\frac{25000}{250} \times 10\right) =$ $1500 + 2500 - \left(\frac{25000}{250} \times 10\right) =$ $1500 - \left(\frac{2500}{250} \times 10\right) =$

Solution

A = 6,250 × 4 = 25,000

O = ₹45

C = 15% × 2.40 = ₹0.36

(a) At present, order size of company is equal to 6,250.

Total annual cost = Purchase cost + Ordering cost + Carrying cost

$$= (25{,}000 \times 2.40) + \left(\frac{25{,}000}{6{,}250} \times 45\right) + \left(\frac{6{,}250}{2} \times 0.36\right) = ₹61{,}305$$

(b) $EOQ = \sqrt{\frac{2 \times A \times O}{C}} = \sqrt{\frac{2 \times 25{,}000 \times 45}{0.36}} = 2{,}500 \text{ units}$

405 Saving

Total annual cost = Purchase cost + Ordering cost + Carrying cost

$$= (25{,}000 \times 2.40) + \left(\frac{25{,}000}{2{,}500} \times 45\right) + \left(\frac{2{,}500}{2} \times 0.36\right) = ₹60{,}900$$

Saving due to EOQ = ₹61,305 – ₹60,900 = ₹405

(c) Re-order level = (Avg. consumption × Avg. lead time) + Safety stock

$$= \left(\frac{25{,}000}{250} \times 10\right) + 500 = 1{,}500 \text{ units}$$

Maximum level = ROL + ROQ – (Min. consumption × Min. lead time)

$$= 1{,}500 + 2{,}500 - \left(\frac{25{,}000}{250} \times 10\right) = 3{,}000 \text{ units}$$

Minimum level = ROL – (Avg. consumption × Avg. lead time)

$$= 1{,}500 - \left(\frac{25{,}000}{250} \times 10\right) = 500 \text{ units}$$

Average level $= \frac{Min.\ level + Max.\ level}{2} = \frac{500 + 3{,}000}{2} = 1{,}750$ units

Prod. = 10000 + 0 − 900 = 9100 → Cons. = 9100 × 2 = 18200 → Purch. = 18200 + 0 − 1000 = 17200

Question – 11	

SK Ltd. produces a product 'SK' using a raw material P. To produce one unit of SK, 2 kg of P is required. As per the sales forecast conducted by the company, it will able to sale 10,000 units of SK in the coming year. The following is the information regarding the raw material P:

$\sqrt{\dfrac{2 \times 17200 \times 720}{13.76\% \times 125}} =$

(i) The Re-order quantity is 200 kg. less than the Economic Order Quantity (EOQ).

50 + 20 = 70

(ii) Maximum consumption per day is 20 kg. more than the average consumption per day. → $\frac{18200}{364} = 50$

(iii) There is an opening stock of 1,000 kg.

(iv) Time required to get the raw materials from the suppliers is 4 to 8 days.

$50 = \frac{70 + \text{Min.}}{2}$

Min. = 30

(v) The purchase price is ₹125 per kg.

There is an opening stock of 900 units of the finished product SK.

The rate of interest charged by bank on Cash Credit facility is 13.76%. C

To place an order company has to incur ₹720 O on paper and documentation work.

From the above information find out the followings in relation to raw material P:

(a) Re-order Quantity → 1200 − 200 = 1000

(b) Maximum Stock level

(c) Minimum Stock level

(d) Calculate the impact on the profitability of the company by not ordering the EOQ.

[Take 364 days for a year]

Solution

Working Notes:

(i) Computation of Annual consumption & Annual Demand for raw material 'P':

Sales forecast of the product 'SK' Less: Opening stock of 'SK'	10,000 units 900 units
Fresh units of 'SK' to be produced	9,100 units
Raw material required to produce 9,100 units of 'SK (9,100 units × 2 kg) Less: Opening Stock of 'P'	18,200 kg. 1,000 kg.
Annual demand for raw material 'P'	17,200 kg

(ii) EOQ = $\sqrt{\dfrac{2 \times 17{,}200 \times 1720}{13.76\%\ of\ 125}}$ = 1,200 kg

(iii) Re- Order level = (Maximum consumption per day × Maximum lead time)

$$= \left(\frac{Annual\ Consumption\ of\ S}{364} + 20kg\right) \times 8 \text{ days} = \left(\frac{18{,}200}{364} + 20kg\right) \times 8days = 560 \text{ kg}$$

(iv) Minimum consumption per day of raw material 'P':

Average Consumption per day = 50 Kg.

Hence, Maximum Consumption per day = 50 kg. + 20 kg. = 70 kg.

So, minimum consumption per day will be

Average Consumption = $\dfrac{Min.Consumption + Max.Consumption}{2}$

50 kg = $\dfrac{Min.Consumption + 70\ kg}{2}$

Min. consumption = 100 kg – 70 kg = 30 kg.

(a) Re-order Quantity = EOQ – 200 kg = 1,200 kg – 200 kg = 1,000 kg

(b) Maximum Stock level = ROL + Re-order Quantity – (Min. consumption × Min. lead time)

= 560 kg. + 1,000 kg. – (30 kg. × 4 days) = 1,440 kg.

(c) Minimum Stock level = ROL – (Average consumption per day × Average lead time)

= 560 kg. – (50 kg. × 6 days) = 260 kg.

(d) Impact on the profitability of the company by not ordering the EOQ.

		When purchasing the ROQ	**When purchasing the EOQ**
I	Order quantity	1,000kg	1,200kg
II	No. of orders a year	$\frac{17,200}{1,000}$ = 17.2 or 18 orders	$\frac{17,200}{1,200}$ = 14.33 or 15 orders
III	Ordering cost	18 orders x ₹720 = ₹12,960	15 orders x ₹720 = ₹10,800
IV	Average inventory	$\frac{1,000}{2}$ = 500kg	$\frac{1,200}{2}$ = 600 kg
V	Carrying cost	500kg x ₹17.2 = ₹8,600	600kg x ₹17.2 = 10,320
VI	Total cost	₹21,560	₹21,120

Extra Cost incurred due to not ordering EOQ = ₹21,560 - ₹21,120 = ₹440

11. ABC Analysis

It stands for always better control analysis. It exercises discriminating control over various items of inventory by classifying them into different categories on the basis of value, quantity, frequency of replacement etc.

Category	% Quantity	% Value	Control
A	10%	70%	High
B	20%	20%	Moderate
C	70%	10%	Low

Question – 12

MM Ltd. has provided the following information about the items in its inventory.

Item Code Number	Units	Unit Cost (₹)	TC	%	Rank
101	25	50	1250	16.67%	A
102	300	01	300	4%	C
103	50	80	4000	53.33%	A
104	75	08	600	8%	B
105	225	02	450	6%	C
106	75	12	900	12%	B
			7500	100%	

MM ltd. has adopted the policy of classifying the items constituting 15% or above to Total Inventory Cost as "A" category, items constituting 6% or less of Total Inventory Cost as "C" category and the remaining items as "B" category. You are required to:

(i) Rank the items on the basis of % of Total Inventory Cost.

(ii) Classify the items into A, B and C categories as per ABC analysis of Inventory Control adopted by MM Ltd.

Solution

Statement of Cost

Item Code Number	Units	Unit Cost (₹)	Total Cost	% of Total Cost	Rank
101	25	50	1,250	16.67%	II
102	300	01	300	4%	VI
103	50	80	4,000	53.33%	I
104	75	08	600	8%	IV
105	225	02	450	6%	V
106	75	12	900	12%	III
	Total		**7,500**	**100%**	

Category	Item Code Number	Total Cost	% of Total Cost
A	103	4,000	53.33%
	101	1,250	16.67%
Total		**5,250**	**70%**
B	106	900	12%
	104	600	8%
Total		**1,500**	**20%**
C	105	450	6%
	102	300	4%
Total		**750**	**10%**

12. Inventory Turnover Ratio (ITR)

- It indicates the number of times inventory has moved out of stores.
- This ratio indicates the efficiency or inefficiency with which inventories are maintained.
- Its purpose is to ensure the blocking of only required minimum funds in inventory.
- A high ratio indicates that goods are fast moving and vice-versa.

ITR for raw material = $\frac{Raw\ material\ consumed}{Average\ raw\ material\ quantity}$ = ___ times

ITR for finished goods = $\frac{Cost\ of\ goods\ sold}{Average\ finished\ goods\ quantity}$ = ___ times

Frequency or Inventory holding period (days) = $\frac{365\ /52\ /12}{ITR}$

Question – 13	

XYZ Ltd uses two types of raw materials – 'Material A' and 'Material B' in the production process and has provided the following data for the year ended on 31st March, 2021:

Particulars	Material A (₹)	Material B (₹)
Opening stock as on 1.04.2020	30,000	32,000
Purchases during the year	90,000	51,000
Closing stock as on 31.02.2021	20,000	14,000

(i) You are required to calculate: Cons. / Avg. $\frac{100000}{25000} \Rightarrow 4$ $\frac{69000}{23000} \Rightarrow 3$ $\frac{360}{4} = 90$

a) The inventory turnover ratio of 'Material A' and ' Material B'

b) The number of days for which the average inventory is held for both materials 'A' and 'B'. → $\frac{360}{3} = 120$

(ii) Based on above calculations, give your comments.

(Assume 360 days in a year)

Solution

(i) Calculation of Inventory Turnover Ratio

Particulars	Material A	Material B
Opening stock	30,000	32,000
Add: Purchases	90,000	51,000
Less: Closing Stock	20,000	14,000
Raw Material Consumed (A)	1,00,000	69,000
Average Stock $\left(\frac{Opening+Closing}{2}\right)$ (B)	$\frac{30,000+20,000}{2}$ = 25,000	$\frac{32,000+14,000}{2}$ = 23,000
Inventory Turnover Ratio (ITR)	$\frac{1,00,000}{25,000}$ = 4 times	$\frac{69,000}{23,000}$ = 3 times
Number of days (360 ÷ ITR)	$\frac{360}{4}$ = 90 days	$\frac{360}{3}$ = 120 days

Supplier — Goods Sent → Fac.

Various Exp.

Supplier ↳ Ignore

Buyer ↳ Add to Cost

13. Choice of Substitute Material

Select the material which has lowest cost per unit of finished goods

	Material A	Material B
Cost per kg	₹20	₹25
Input-output ratio	200%	120%
Cost per unit of output	₹40	₹30

14. Landing Cost of Material or Valuation of Material

Items	Treatment
Trade Discount	Deduct if not already deducted
Cash Discount	Ignore
Subsidy/Grant/Incentive	Deduct from material cost
Road tax/ Toll tax/	Add to material cost
IGST/CGST/SGST	
(A) If ITC available	Ignore
(B) If ITC not available	Add to material cost
Custom Duty	Add to material cost
Penalty / Fine / Demurrage Abnormal	Ignore
Insurance	Add to material cost
Commission	Add to material cost
Freight Inward	Add to material cost
Container Cost	Add to material cost
Return value of container	Deduct the returnable value of container from material cost Show as Net value
Shortage	
(A) Normal	Deduct quantity of normal loss from total quantity
(B) Abnormal	Transfer to P&L account

- **GST is payable on net purchase price** i.e. List price – Trade Discount
- **Distribution of Freight or similar items** – On the basis of quantity of material
- **Distribution of GST, Custom duty or similar items** – On the basis of value of material

No GST Input

Question – 14

SK & Co., an unregistered supplier under GST, purchased material from PK Ltd. which is registered under GST. The following information is available for one lot of 5,000 units of material purchased:

Listed price of one lot	₹2,50,000
Trade discount	@ 10% on listed price
CGST and SGST (Credit Not available)	12% (6% CGST + 6% SGST)
Cash discount (Ignore)	@ 10%
(Will be given only if payment is made within 30 days.)	
Toll Tax paid	₹5,000
Freight and Insurance	₹17,000
Demurrage paid to transporter (Ab.)	₹5,000
Commission and brokerage on purchases	₹10,000
Amount deposited for returnable containers	₹30,000
Amount of refund on returning the container	₹20,000
Other Expenses	@ 2% of total cost

₹10000

(+) oth. Exp. @2%

TC (294000 ÷ 98%)

Qty. (5000 – 20%)

250000
(25000)
225000
27000
252000
5000
17000
10000
10000
294000
6000
300000
4000
75

20% of material shortage is due to normal reasons. The payment to the supplier was made within 21 days of the purchases. You are required to calculate cost per unit of material purchased by SK & Co.

Solution

Statement of calculation of cost per unit

Particulars	Amount (₹)
Listed price of materials (on lot)	2,50,000
Less: Trade discount @ 10% on listed price	(25,000)
	2,25,000
Add: CGST @ 6% of 2,25,000	13,500
Add: SGST @ 6% of 2,25,000	13,500
	2,52,000
Add: toll tax	5,000
Add: Freight and insurance	17,000
Add: Commission and brokerage paid	10,000
Add: Cost of refundable containers (30,000 – 20,000)	10,000
	2,94,000
Add: Other expenses (2,94,000 ÷ 98%)	6,000
Total cost of material (A)	3,00,000
Total quantity of material in one lot	5,000 units
Less: Normal loss @20% of 5,000	1,000 units
Net quantity of material (B)	4,000 units
Material cost per unit (A ÷ B)	75

Note:

(a) GST is payable on net price i.e. listed price less trade discount

(b) Cash discounts is treated as interest and finance cost, hence it is ignored.

(c) Demurrage is penalty imposed by the transporter for delay in uploading or off-loading of materials. It is an abnormal cost and thus, not included.

Question – 15	

An invoice in respect of a consignment of chemicals A and B provide the following information:

	(₹)
Chemical A: 10,000 kgs at ₹10 per kg	1,00,000
Chemical B: 8,000 kgs at ₹13 per kg	1,04,000
Basic custom duty @10% (credit is not allowed) [100 : 104]	20,400
Railway freight [10 : 8]	3,840
	2,28,240

A shortage of 500 kg in chemical A and 320 kg in chemical B is noticed due to normal breakages. You are required to compute the rate per kg of each chemical, assuming a provision of 2% for further deterioration.

Solution

Statement of Cost

Particulars	**Chemical A**	**Chemical B**
Purchase price	1,00,000	1,04,000
Add: Basic custom duty @10%	10,000	10,400
(+) Railway freight	2,133	1,707
(₹3,840 in ratio of 5:4 i.e. quantity purchased)		
Total Cost (A)	1,12,133	1,16,107
Quantity Purchased	10,000	8,000
(-) Normal breakage	(500)	(320)
	9,500	7,680
(-) Provision for detonation @ 2%	(190) 2%	(153.6) 2%
Net Quantity (B)	9,310	7,526.4
Total cost per kg (A ÷ B)	12.04	15.43

Question – 16	

XYZ Enterprises manufactures different types of beverages. The monthly demand pattern for these beverages is as follows:

- Orange Juice: 60,000 litres → A = 60000×12×3 = 21.6L
- Apple Juice: 20,000 litres → A = 20000×12×6 = 14.40L
- Grape Juice: 5,000 litres → A = 5000×12×5 = 3L

To produce one litre of Apple Juice, Orange Juice, and Grape Juice, 6 kg of apples, 3 kg of oranges, and 5 kg of grapes are required, respectively. There is no opening or closing stock of raw materials.

XYZ Enterprises can source the raw materials either from local farmers (import in case or grapes) or from wholesale suppliers. The company has the flexibility to purchase any quantity of materials from the wholesale suppliers, but when buying from the farmers or in case of import, it must adhere to a minimum quantity that is specified for each type of material. Below is a summary of the purchasing conditions:

Material	Wholesale Supplier	Farmers
Oranges	Minimum purchase: Any quantity	7,20,000 kg
	Price per kg: ₹18.00 +0.2+0.04=18.24	Price per kg: ₹15.00 +0.1+0.04=15.14
	Transportation: ₹5,000 0	Transportation: ₹20,000 0 → 21500
	-	Sorting and piling cost: ₹1,500 0
PC	Loading cost: ₹10/50 kg = 0.2	Loading cost: ₹5/50 kg = 0.1
	Unloading cost: ₹2/50 kg = 0.04	Unloading cost: ₹2/50 kg = 0.04
Apples	Minimum purchase: Any quantity	3,60,000 kg
	Price per kg: ₹12.00 + 0.2+0.04=12.24	Price per kg: ₹10.00 +0.06+ 0.04=10.10
	Transportation: ₹11,000 0	Transportation: ₹15,000 0 → 16200
	-	Sorting and piling cost: ₹1,200 0
	Loading cost: ₹10/50 kg = 0.2	Loading cost: ₹3/50 kg = 0.06
	Unloading cost: ₹2/50 kg = 0.04	Unloading cost: ₹2/50 kg = 0.04
	Wholesaler Supplier	**Import**
Grapes	Minimum purchase: Any quantity	1,50,000 kg
	Price per kg: ₹32.00 +0.2+0.04=32.24	Price per kg: ₹25.00 +0.5+0.04+(25×8%) =27.54
	Transportation: ₹3,000 0	Transportation: ₹15,000 0
	Loading cost: ₹10/50 kg = 0.2	Loading cost: ₹25/50 kg = 0.50
	Unloading cost: ₹2/50 kg = 0.04	Unloading cost: ₹2/50 kg = 0.04

Transportation and Sorting and piling cost is given for per purchase order.
8% import duty is applicable is purchase of grapes for which ITC is not available.
XYZ Enterprises is also paying an annual interest rate of 15% on cash credit facility and ₹1,000 per 1,000 kg of rent to its warehouse. → $\frac{1000}{1000}$ = 1

You are required to:

1. Calculate the total purchase cost for each material (Apple, Orange, and Grape) under the following two options:
   (a) Purchase from Wholesale Supplier
   (b) Purchase from Farmers/Import
2. Calculate the Economic Order Quantity (EOQ) for each material under both options.
3. Provide a Recommendation on the best purchase option for the "Grapes" material if minimum purchase quantity for import is 1,50,000 kg.

Solution

(a) Calculation of purchase cost per kg of Materials

	Wholesale Market (₹)	Farmers (₹)
Orange:		
Purchase price Add: Loading cost (₹10 ÷ 50kg) (₹5÷ 50kg)	18.00 0.20	15.00 0.10

Add: Unloading cost (₹2 ÷ 50kg)	0.04	0.04
Total	18.24	15.14
Apple:		
Purchase price Add: Loading cost (₹10 ÷ 50kg) (₹3 ÷ 50kg) Add: Unloading cost (₹2 ÷ 50kg)	12.00 0.20 0.04	10.0 0.06 0.04
Total	12.24	10.10
Grape:		
Purchase price Add: Import duty @ 8% Add: Loading cost (₹10 ÷ 50kg) (₹25 ÷ 50kg) Add: Unloading cost (₹2 ÷ 50kg)	32.00 - 0.20 0.04	25.00 2.00 0.50 0.04
Total	32.24	27.54

(b) Calculation of EOQ

Commodity	Annual requirement (Kg)
Orange (60,000 ltr. × 3kg × 12 months)	21,60,000
Apple (20,000 ltr. × 6kg × 12 months)	14,40,000
Grape (5,000 ltr. × 5kg × 12 months)	3,00,000

Cost per order (O)

Commodity	Wholesale market (₹)	Farmers (₹)
Orange:		
Transportation cost Sorting and piling cost	5,000 -	20,000 1,500
Total	5,000	21,500
Apple:		
Transportation cost Sorting and piling cost	11,000 -	15,000 1,200
Total	11,000	16,200
Grape:		
Transportation cost Sorting and piling cost	3,000 -	15,000 -
Total	3,000	15,000

Carrying cost per Kg per annum (C)

Commodity	Wholesale market (₹)	Farmers (₹)
Orange:		
Interest on cash credit Warehouse rent (₹100 ÷ 100 kg)	(₹18.24 × 15%) 2.736 1.0000	(₹15.14 × 15%) 2.271 1.0000
Total	3.736	3.271
Apple:		
Interest on cash credit Warehouse rent (₹100 ÷ 100 kg)	(₹12.24 × 15%) 1.836 1.0000	(₹10.10 × 15%) 1.515 1.0000
Total	2.836	2.515

Grape:		
Interest on cash credit Warehouse rent (₹100 ÷ 100 kg)	(₹32.24 × 15%) 4.836 1.0000	(₹27.54 × 15%) 4.131 1.0000
Total	5.836	5.131

Calculation of EOQ:

Commodity	Wholesale market (₹)	Farmers (₹)
Orange	$\sqrt{\frac{2\times21,60,000\times5,000}{3.736}}$ = 76,036.72	$\sqrt{\frac{2\times21,60,000\times21,500}{3.271}}$ = 1,68,508.11
Apple	$\sqrt{\frac{2\times14,40,000\times11,000}{2.836}}$ = 1,05,691.36	$\sqrt{\frac{2\times14,40,000\times16,200}{2.515}}$ =1,07,798.07
Grape	$\sqrt{\frac{2\times3,00,000\times3,000}{5.836}}$ = 17,562.19	$\sqrt{\frac{2\times3,00,000\times15,000}{5.131}}$ = 41,881.31

(c) Selection of best purchase option for the purchase of Grapes:

	Wholesale Market (₹)	Farmer (₹)
Annual requirement (A)	3,00,000	3,00,000
Order quantity (Q)	17,562.19	1,50,000
No. of orders (A ÷ Q)	17.08 or 18	2
Average quantity (Q ÷ 2)	8,781	75,000
Ordering cost (₹) (I)	54,000 (18 orders × 3,000)	30,000 (2 order × 15,000)
Carrying cost (₹) (II)	51,246 (8,781 kg × 5.836)	3,84,825 (75,000 kg × 5.131)
Purchase cost (₹) (III)	96,72,000 (3,00,000 kg × 32.24)	82,62,000 (3,00,000 kg × 27.54)
Total cost (I + II + III)	97,74,245.92	86,76,825

Purchasing Grapes direct from the farmers is the best purchase option for the XYZ enterprises.

Prod. = (30000×12) + 0 − 0 = 3.6L ⟶ Cons. = 3.60L × $\frac{5}{3}$ = 6L → K = 6L × 3/5 = 3.60L → Purch.; M = 6L × 2/5 = 2.40L → Purch.

Question – 17	

SK, a small scale manufacturer, produces a product S by using two raw materials K and M in the ratio of 3:2. Material K is perishable in nature and if not used within 5 days of purchase it becomes obsolete. Material M is durable in nature and can be used even after one year. The company has estimated a sales volume of 30,000 kg. for the month of October and expects that the trend will continue for the entire year. The ratio of input and output is 5:3. The purchase price of per kilogram of raw material K and M is ₹15 and ₹22 respectively exclusive of taxes. Material K can be purchased from the local market within I to 2 days period. On the other hand, Material M is purchased from neighboring state and it takes 2 to 4 days to receive the material in the store.

To place an order the company has to incur an administrative cost of ₹120. Carrying cost for Material K and M is 15% and 5% respectively.

Present OS

At present Material K is purchased in a lot of 8,000 kg. to avail 10% discount on market price. SGST & CGST applicable for material K is 4% (credit available) and IGST on Material M is 2% (credit not available). Company works for 25 days in a month and production is carried out evenly.

K = $\sqrt{\frac{2\times360000\times120}{15\%\times15}}$ = M = $\sqrt{\frac{2\times240000\times120}{5\%\times(22.44)}}$ =

You are required to calculate:

(a) Economic Order Quantity (EOQ) for each material;

(b) Maximum stock level for Material K; → $\left(\frac{3.60L}{25\times12}\times2\right)+8000-\left(\frac{3.60L}{25\times12}\times1\right)$ = 9200 (or) $\left(\frac{3.60L}{25\times12}\times5\right)$ = 6000

(c) Calculate saving/loss in Material K if purchase quantity equals to EOQ.

6000

Solution

(a) Annual sales = 30,000 × 12 = 3,60,000

Annual raw material requirement = 3,60,000 × (5/3) = 6,00,000 kg

Annual raw material requirement (A) of material K = 6,00,000 × (3/5) = 3,60,000 kg

Annual raw material requirement (A) of material M = 6,00,000 – 3,60,000 = 2,40,000 kg

Cost per order (O) = ₹120

C for material K = 15% × 15 = ₹2.25

C for material M = 5% × (22 + 2%) = ₹1.122

EOQ for material K = $\sqrt{\frac{2\times A\times O}{C}} = \sqrt{\frac{2\times3,60,000\times120}{2.25}}$ = 6,197 kg

EOQ for material M = $\sqrt{\frac{2\times A\times O}{C}} = \sqrt{\frac{2\times2,40,000\times120}{1.122}}$ = 7,165 kg

(b) Maximum level for material K = ROL + ROQ – (Avg. consumption × Avg. lead time)

= (Max. consumption × Max. time) + ROQ – (Avg. consumption × Avg. time)

= $\left(\frac{3,60,000}{25\times12}\times2\right)$ + 8,000 - $\left(\frac{3,60,000}{25\times12}\times1\right)$ = 9,200 kg

Also, since material K is perishable in nature and will become obsolete after 5 days,

∴ Maximum level = $\left(\frac{3,60,000}{25\times12}\times5\right)$ = 6,000 kg

So maximum level will be minimum of the two values i.e. 6,000 kg and 9,000 kg.

∴ Maximum level for material A = 6,000 kg

(c) No. of order = 60 **Statement of Evaluation of Offer**

Costs	Order Size = 6,197	Order Size = 8,000
Purchase Cost	3,60,000 × 15 = 54,00,000	3,60,000 × (15-10%) = 48,60,000
Ordering Cost	$\frac{3,60,000}{6,000} \times 120 = 7,200$	$\frac{3,60,000}{6,000} \times 120 = 7,200$
Carrying Cost	$\frac{6,197}{2} \times 15\% \times 15 = 6,972$	$\frac{8,000}{2} \times 15\% \times 13.50 = 8,100$
Loss due to perishable nature	197 × 60 × 15 = $\boxed{1,77,300}$	2,000 × 60 × 13.50 = $\boxed{16,20,000}$
Total Cost	55,91,472	64,95,300

Savings in cost due to EOQ = ₹64,95,300 – ₹55,91,472 = ₹9,03,828

15. Safety Stock Determination

- It occurs when an inventory item could not be supplied due to insufficient stock in the store.
- It leads to both financial and non-financial loss to company.
- If high safety stock is maintained than it will lead to high carrying cost whereas in case of low or zero safety stock, it will lead to high stock out cost.
- Thus, it is a trade-off between the carrying cost and stock-out cost.

Carrying cost of safety stock = Safety stock unit × Carrying cost per unit per annum

Annual Stock out cost = Annual stock out units × Stock out cost per unit

Question – 18

M/s SK Ltd trades in chairs. It stocks sufficient quantity of chairs of almost every variety. In year end, the report of sales manager revealed that M/s SK experienced stock-out of chairs. The stock-out data is as follows:

Stock-out of chairs	No. of times	% or Prob.
100	2	0.02
80	5	0.05
50	10	0.10
20	20	0.20
10	30	0.30
0	33	0.33
	100	1

M/s SK loses ₹150 per unit due to stock-out and spends ₹50 per unit on carrying of inventory. Determine optimum safest stock level.

Solution

Computation of probability of stock out

Stock-out (units)	100	80	50	20	10	0	Total
No. of times	2	5	10	20	30	33	100
Probability	0.02	0.05	0.10	0.20	0.30	0.33	1.00

Statement showing determination of Optimal Stock

Safety Stock Units	Stock-out units	Prob.	Expected annual stock out units	Expected annual stock out costs	Annual holding cost	Total annual expected cost
100	0	0	0	0	5,000	5,000
80	20	0.02	0.4	60	4,000	4,060
50	50 30	0.02 0.05	1.0 1.5 2.5	150 225 375	2,500	2,875

20	80 60 30	0.02 0.05 0.10	1.6 3 3 7.6	240 450 450 1,140	1,000	2,140
10	90 70 40 10	0.02 0.05 0.10 0.20	1.8 3.5 4.0 2.0 11.3	270 525 600 300 1,695	500	2,195
0	100 80 50 20 10	0.02 0.05 0.10 0.20 0.30	2 4 5 4 3 18	300 600 750 600 450 2,700	0	2,700

It is recommended to maintain safety stock level of 20 units at which total cost is least i.e. ₹2,140.

16. Material Records

It can be done in two ways i.e. Perpetual system and Periodic system.

17. Preparation of Stores Ledger

(A) Material return from factory or production to stores

- Show as receipt at the price at which originally issued
- If price not known than at recent issue rate.
- To be issued first in FIFO or LIFO method

(B) Material return by stores to supplier or vendor

- Show as issued in stores ledger at the price at which originally purchased
- If original price not known than at recent issue rate.

(C) Transfer from one job to another

- No entry in stores ledger

(D) **In case of normal loss**, show as issue in quantity column only and thus price of balance quantity increases.

(E) **In case of abnormal loss,** show as issue as per the method prevailing and transfer the same to costing P&L account.

(F) **Material Consumed** = Issue of material (option-1)

Material consumed = Opening Stock + Purchases – Purchase return – Abnormal loss – Closing stock (option-2)

Question – 19	**[SM]**

SK Ltd. furnishes the following store transactions for September:

1 Sep	Opening balance	25 units value ₹162.50
4 Sep	Issues Req. No. 85	8 units
6 Sep	Receipts from B & Co. GRN No. 26	50 units @ ₹5.75 per unit
7 Sep	Issues Req. No. 97	12 units
10 Sep	Return to B & Co.	10 units @ 5.75
12 Sep	Issues Req. No. 108	15 units
13 Sep	Issues Req. No. 110	20 units
15 Sep	Receipts from M & Co. GRN No. 33	25 units @ ₹6.10 per unit
17 Sep	Issues Req. No. 121	10 units
19 Sep	Received replacement from B & Co. GRN No. 38	10 units @ 5.75
20 Sep	Returned from department, material of M & CO. MRR No. 4	5 units
22 Sep	Transfer from Job 182 to Job 187 in the dept. MTR 6	5 units ×
26 Sep	Issues Req. No. 146	10 units

29 Sep Transfer from Dept. A to Dept. B MTR 10 5 units
30 Sep Shortage in stock taking (Ab.) 2 units

Write up the priced stores ledger on FIFO method and discuss how would you treat the shortage in stock taking

Solution

Date	GRN / MRN	Receipt			Requisition No.	Issue			Balance		
		Qty	Rate	Amount		Qty	Rate	Amount	Qty	Rate	Amount
1 Sep	-	-	-	-	-	-	-	-	25	6.50	162.5
4 Sep	-	-	-	-	85	8	6.5	52	17	6.50	110.5
6 Sep	26	50	5.75	287.50	-	-	-	-	17 50	6.50 5.75	162.50 287.50
7 Sep	-	-	-	-	97	12	6.50	78	5 50	6.50 5.75	32.50 287.50
10 Sep	-	-	-	-	-	10	5.75	57.50	5 40	6.50 5.75	32.50 230
12 Sep	-	-	-	-	108	5 10	6.50 5.75	32.50 57.50	30	5.75	172.50
13 Sep	-	-	-	-	110	20	5.75	115	10	5.75	57.50
15 Sep	33	25	6.10	152.50	-	-	-	-	10 25	5.75 6.10	57.50 152.50
17 Sep	-	-	-	-	121	10	5.75	57.50	25	6.10	152.50
19 Sep	38	10	5.75	57.50	-	-	-	-	25 10	6.10 5.75	152.50 57.50
20 Sep	4	5	5.75 (AB.)	28.75	-	-	-	-	5 25 10	5.75 6.10 5.75	28.75 152.50 57.50
26 Sep	-	-	-	-	146	5 5	5.75 6.10	28.75 30.50	20 10	6.10 5.75	122 57.50
30 Sep	-	-	-	-	Shortage	2	6.10	12.20	18 10	6.10 5.75	109.80 57.50

Working Notes:

(a) The material received as replacement form vendor is treated as fresh supply

(b) In the absence of any information, the price of the material returned from a user department on 20 Sep has been taken at the price of the latest issue made on 17 Sep. In FIFO method, physical flow of the material is irrelevant and issue price is based on first in first out.

(c) The issue of material on 26 Sep is made out of the material received from a user department on 20 Sep.

(d) The entries for transfer of material from one job to another on 22 Sep and 29 Sep do not affect the store ledger. Hence no entries are passed in its respective.

(e) The material found short as a result of stock taking has been written off at relevant issue price.